CAMBRIDGE
International Education

# Cambridge O Level

CANDIDATE NAME

CENTRE NUMBER

CANDIDATE NUMBER

*0312918659*

**MATHEMATICS (SYLLABUS D)** **4024/22**

Paper 2 Calculator **May/June 2026**

**2 hours**

You must answer on the question paper.

You will need: Geometrical instruments

**INSTRUCTIONS**

- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen. Do **not** use correction fluid or tape.
- Do **not** write on any bar codes.
- You should use a scientific calculator where appropriate.
- You may use tracing paper.
- You must show all necessary working clearly.
- Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place for angles in degrees, unless a different level of accuracy is specified in the question.
- For $\pi$, use either your calculator value or 3.142.

**INFORMATION**

- The total mark for this paper is 100.
- The number of marks for each question or part question is shown in brackets [ ].

This document has **20** pages.

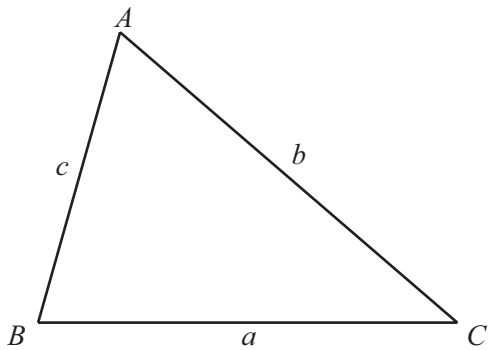
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**[Turn over**

## List of formulas

| | |
|---|---|
| Area, $A$, of triangle, base $b$, height $h$. | $A = \frac{1}{2}bh$ |
| Area, $A$, of circle of radius $r$. | $A = \pi r^2$ |
| Circumference, $C$, of circle of radius $r$. | $C = 2\pi r$ |
| Curved surface area, $A$, of cylinder of radius $r$, height $h$. | $A = 2\pi rh$ |
| Curved surface area, $A$, of cone of radius $r$, sloping edge $l$. | $A = \pi rl$ |
| Surface area, $A$, of sphere of radius $r$. | $A = 4\pi r^2$ |
| Volume, $V$, of prism, cross-sectional area $A$, length $l$. | $V = Al$ |
| Volume, $V$, of pyramid, base area $A$, height $h$. | $V = \frac{1}{3}Ah$ |
| Volume, $V$, of cylinder of radius $r$, height $h$. | $V = \pi r^2 h$ |
| Volume, $V$, of cone of radius $r$, height $h$. | $V = \frac{1}{3}\pi r^2 h$ |
| Volume, $V$, of sphere of radius $r$. | $V = \frac{4}{3}\pi r^3$ |
| For the equation $ax^2 + bx + c = 0$, where $a \neq 0$ | $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$ |

For the triangle shown,




$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

$$a^2 = b^2 + c^2 - 2bc\cos A$$

$$\text{Area} = \frac{1}{2}ab\sin C$$

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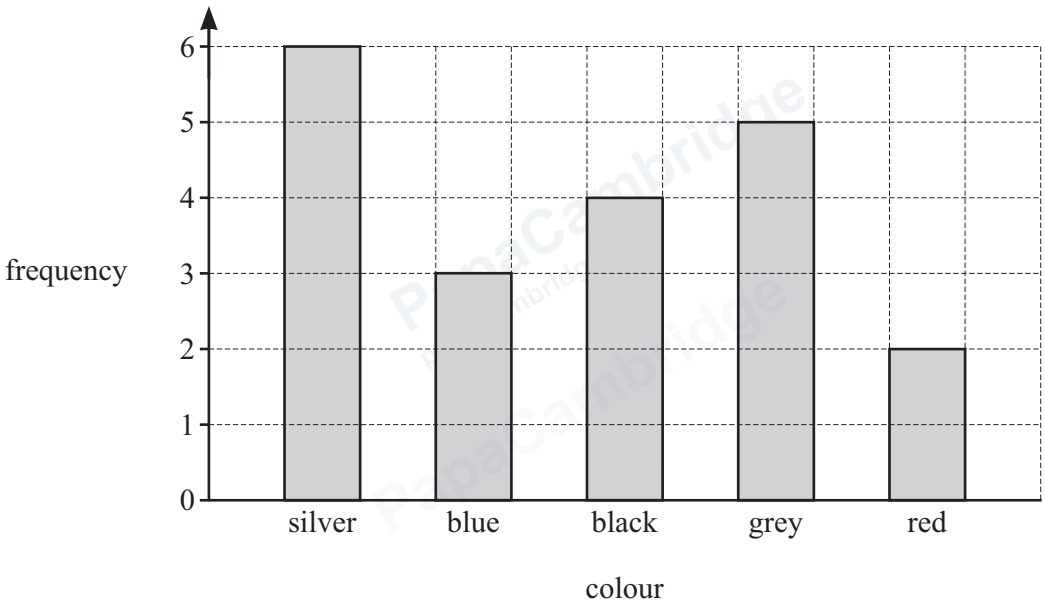
**1** Write these numbers in order of size, starting with the smallest.

$\frac{2}{5}$ 0.42 38%

..... , ..... , ..... [1]

*smallest*

**2** Maison records the colours of all the cars in a car park.
The results are shown in the bar chart.




**(a)** Write down the colour of car that is recorded the least number of times.

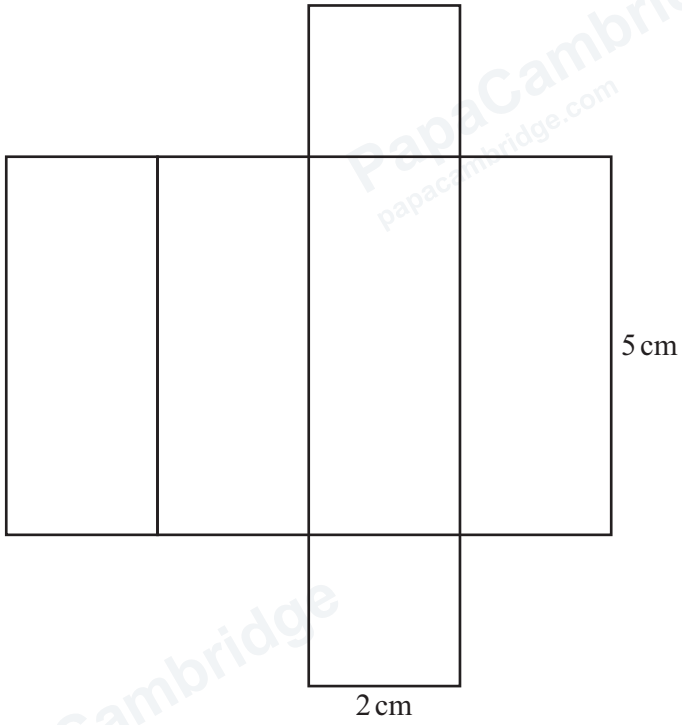
..... [1]

**(b)** Find the total number of cars in the car park.

..... [1]

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**3** The diagram shows the net of a cuboid made from four congruent rectangles and two congruent squares.




Find the total length of the edges of the cuboid.

..... cm [2]

**4** Calculate.

$$\frac{2.5^4}{3.09 + \sqrt{16.98}}$$

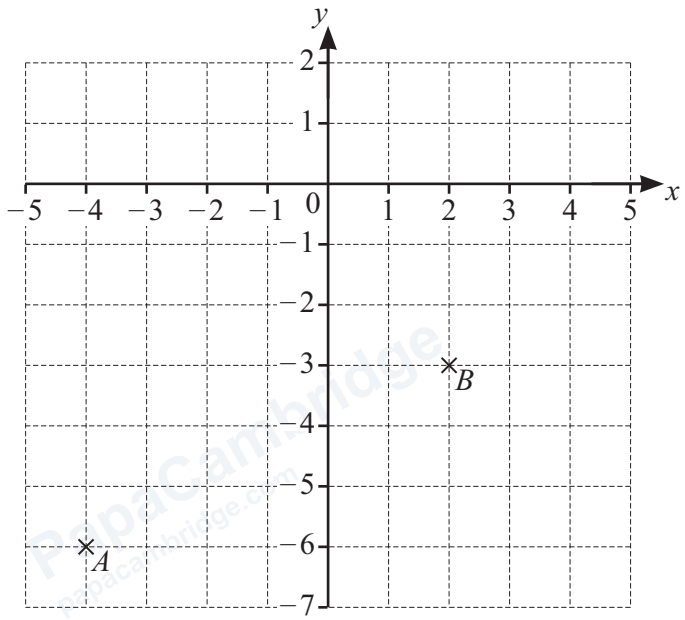
Give your answer correct to 2 decimal places.

..... [2]

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**5** Point $A$ and point $B$ are plotted on the grid.




**(a)** Write down the coordinates of point $B$.

(..... , .....) [1]

**(b)** Work out the gradient of $AB$.

..... [2]

**(c)** Find the coordinates of the midpoint of the line $AB$.

(..... , .....) [1]

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**6** Sara and Bill share some money in the ratio 5 : 6.
Sara receives $60.

**(a)** Calculate the amount Bill receives.

$ ..... [2]

**(b)** Sara uses her $60 to buy books.
Each book costs $7.

Find the maximum number of books she can buy.

..... [2]

**7** **(a)** $\mathbf{c} = \begin{pmatrix} 1 \\ 5 \end{pmatrix}$ $\mathbf{d} = \begin{pmatrix} -2 \\ 3 \end{pmatrix}$

Write $\mathbf{c} - 2\mathbf{d}$ as a column vector.

$\begin{pmatrix} \quad \\ \quad \end{pmatrix}$ [2]

**(b)** Point $P$ has coordinates (1, 5) and $\overrightarrow{PQ} = \begin{pmatrix} -4 \\ 10 \end{pmatrix}$.

Find the coordinates of point $Q$.

( ..... , ..... ) [2]

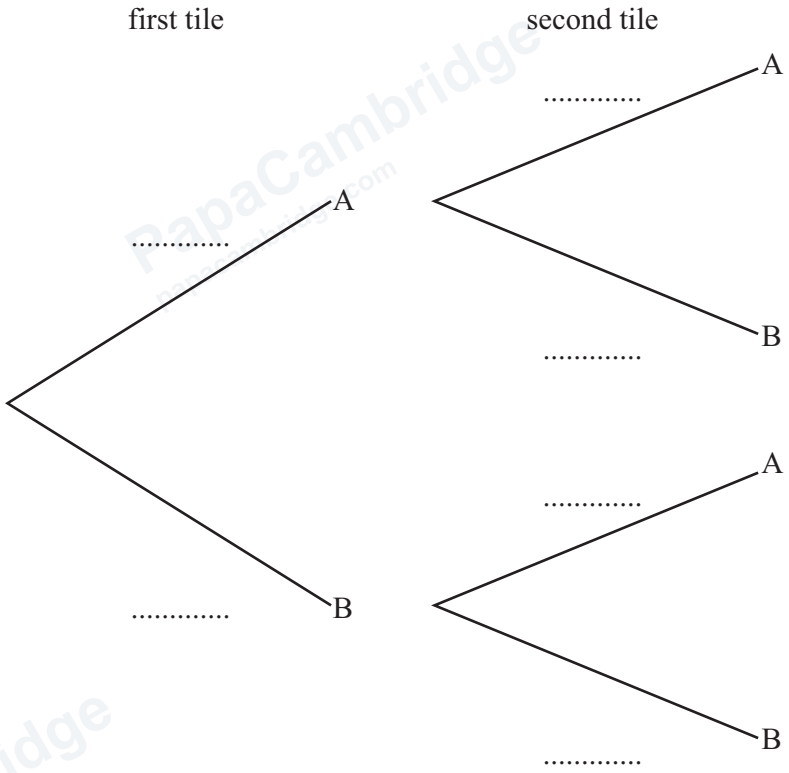
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**8** A bag contains 7 tiles labelled A and 5 tiles labelled B.
A tile is chosen at random from the bag and is then replaced.
A second tile is then chosen at random from the bag.

**(a)** Complete the tree diagram.




[2]

**(b)** Work out the probability that both tiles are labelled A.

..... [1]

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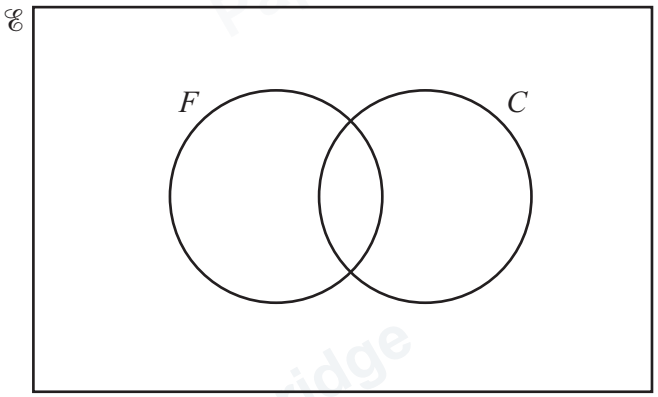
**9** Amira walks from her home to her friend's house.
She leaves home at 09 35.
She walks a total of 8 km at an average speed of 5 km / h.

Work out the time she arrives at her friend's house.

..... [3]

**10** $\mathscr{E} = \{x : x \text{ is an integer and } 1 \leqslant x \leqslant 9\}$
$F = \{x : x \text{ is a factor of } 6\}$
$C = \{x : x \text{ is a cube number}\}$

**(a)** Complete the Venn diagram.




[2]

**(b)** Find $n(F \cup C)'$.

..... [1]

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**11** In a sale, a shop reduces all prices by 32%.
Before the sale, a dress costs $56.

Work out the sale price of the dress.

$ ..... [2]

**12** Solve.

**(a)** $5k - 2 = 12$

$k =$ ..... [2]

**(b)** $5(2y - 1) = 4(3y + 1)$

$y =$ ..... [3]

**13** $f(x) = 2x + 3$ $g(x) = \frac{1}{x-2}$, $x \neq 2$

**(a)** Find $f(-4)$.

..... [1]

**(b)** Find $gf(1)$.

..... [2]

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**14** The population of a village is 1260.
The population grows exponentially at a rate of 1.7% per year.

Calculate the increase in the population at the end of 3 years.

..... [3]

**15** Simplify.

**(a)** $a^5 \times a^3$

..... [1]

**(b)** $b \div b^6$

..... [1]

**(c)** $\left(\frac{4x}{y^2}\right)^3$

..... [2]

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**16** The table gives information about the speeds of 60 cars travelling along a road.

| **speed ($s$ km/h)** | $20 < s \leqslant 24$ | $24 < s \leqslant 40$ | $40 < s \leqslant 58$ | $58 < s \leqslant 70$ |
|---|---|---|---|---|
| **frequency** | 4 | 22 | 28 | 6 |

Calculate an estimate of the mean.

..... km/h [4]

**17** **(a)** $f$ is inversely proportional to the square of $h$.

When $f = -\frac{3}{4}$, $h = 2$.

Find $f$ when $h = -5$.

$f =$ ..... [3]

**(b)** $c$ is directly proportional to $d^2$.

Complete the statement.

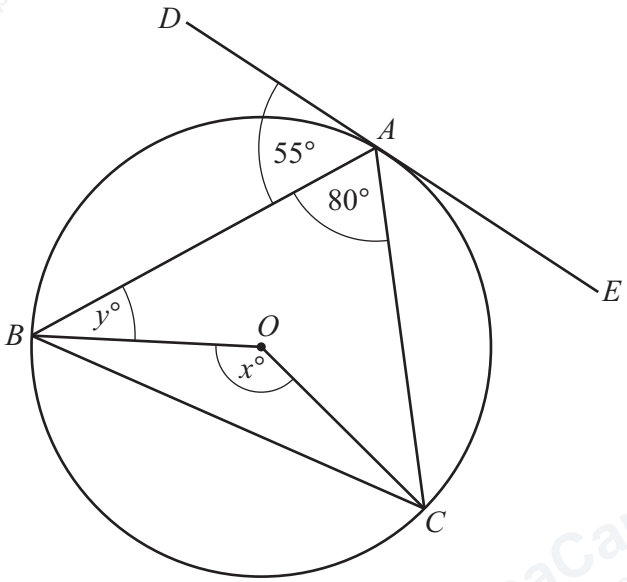
If $d$ is multiplied by 4, then $c$ is multiplied by ..... [1]

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18




NOT TO SCALE

$A$, $B$ and $C$ lie on a circle, centre $O$.
$DE$ is a tangent to the circle at $A$.
Angle $BAC = 80°$ and angle $BAD = 55°$.

**(a)** Find the value of $x$.

Give a geometrical reason for your answer.

$x =$ ..... because .....

..... [2]

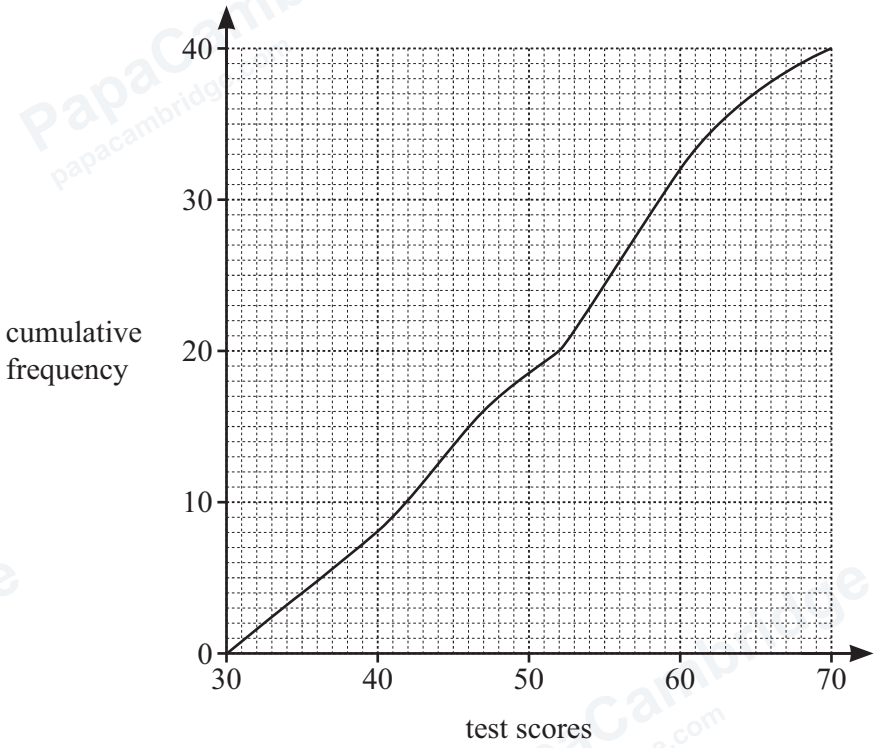
**(b)** Find the value of $y$.

$y =$ ..... [2]

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**19** The cumulative frequency diagram shows information about the test scores for a group of students.




**(a)** Use the diagram to find an estimate for the median score.

..... [1]

**(b)** In this group, 60% of the students pass the test.

Use the diagram to find the minimum score needed to pass the test.

..... [2]

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**20** The area of a rectangle is $69.5\,m^2$, correct to the nearest $0.5\,m^2$.
The width of the rectangle is 3.8 m, correct to the nearest 0.1 m.

Calculate the upper bound of the length of the rectangle.

..... m [3]

**21** Simplify.

**(a)** $\frac{8}{32x}$

..... [1]

**(b)** $\frac{2f^2 - 8f}{f^2 - 16}$

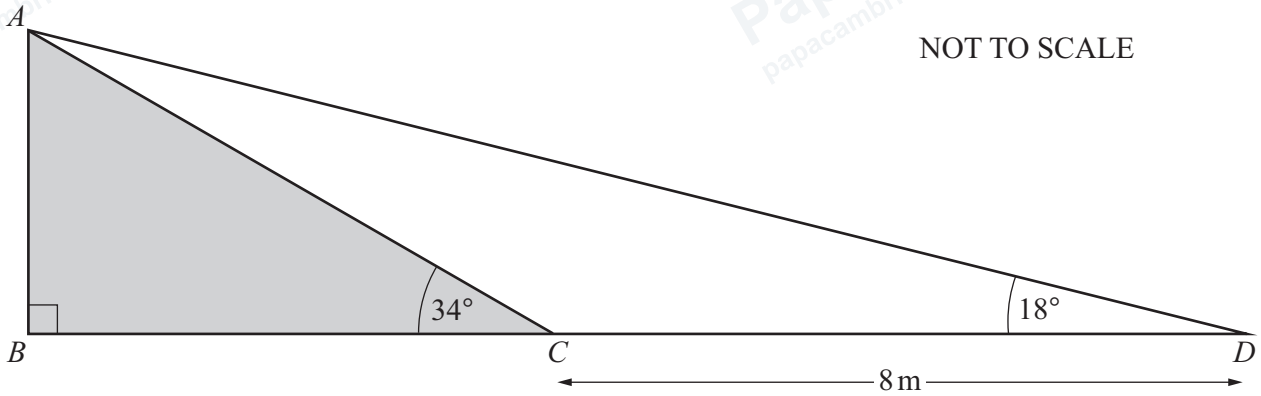
..... [3]

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**22** The diagram shows a triangular garden $ABD$.
A path from $A$ meets $BD$ at point $C$.




$CD = 8\,\text{m}$, angle $ADC = 18°$, angle $ACB = 34°$ and angle $ABC = 90°$.

**(a)** Calculate the area of triangle $ABC$.

..... $\text{m}^2$ [6]

**(b)** The shaded area $ABC$ is covered with grass seed.
A box of grass seed costs \$6.
Each box contains enough grass seed to cover an area of $5\,\text{m}^2$.

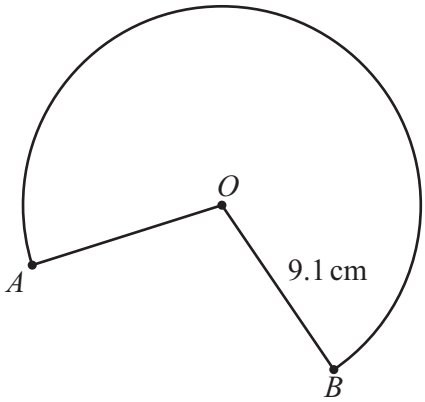
Calculate the total cost of the boxes needed to cover this area with seed.

\$ ..... [2]

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**23**



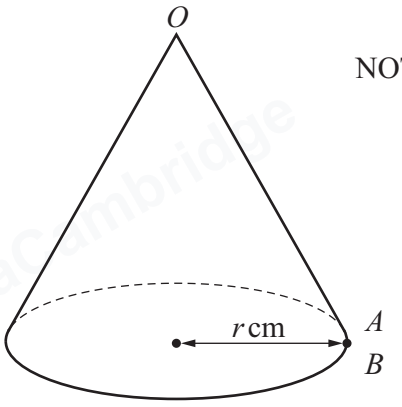

NOT TO SCALE

The diagram shows a sector of a circle with centre $O$ and radius 9.1 cm.
The length of the major arc $AB$ is 44 cm.

**(a)** Find the angle of the major sector $AOB$.

..... [3]

**(b)** Jamila joins $OA$ to $OB$ to form the curved surface of a cone.




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**(i)** Show that the radius, $r$, of the cone is 7.0 cm, correct to 1 decimal place.

[2]

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**(ii)** Calculate the volume of the cone.

..... $cm^3$ [4]

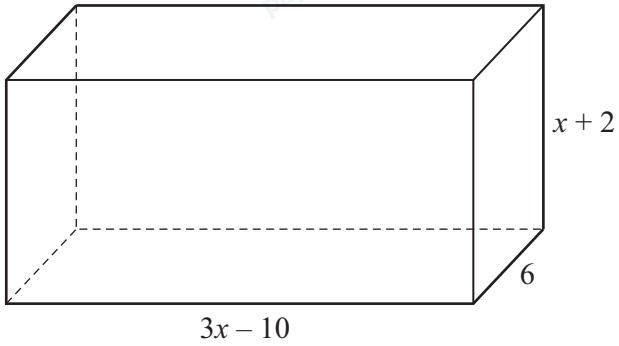
**(iii)** Oliver has a cone that is mathematically similar to Jamila's cone.
The volume of Oliver's cone is $2387\,cm^3$.

Calculate the radius of Oliver's cone.

..... cm [3]

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**24** The diagram shows a solid cuboid.
All the lengths are in centimetres.




NOT TO SCALE

**(a)** The volume of the cuboid is $720\,\text{cm}^3$.

Show that $3x^2 - 4x - 140 = 0$.

[3]

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**(b)** Use the quadratic formula to solve $3x^2 - 4x - 140 = 0$.
Show all your working and give your answers correct to 2 decimal places.

$x =$ ..... or $x =$ ..... [3]

**(c)** Calculate the area of the top face of the cuboid.

..... $cm^2$ [2]

**Question 25 is on the back page.**



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**25** **(a)** Write $2x^2 - 8x + 7$ in the form $a(x-b)^2 + c$.

..... [3]

**(b)** Write down the coordinates of the turning point of the graph of $y = 2x^2 - 8x + 7$.

(..... , .....) [1]

**(c)** The equation $2x^2 - 8x + 7 = k$ has two solutions.

Complete the inequality for $k$.

$k >$ ..... [1]

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